CAMERON GORDON

Boston, MA • (858) 413-6322 • Cameron.Gordon.Eng@gmail.com • linkedin.com/in/gordonca • Active Secret Clearance

Electrical Engineer (M.S. ECE) designing space-grade flight computers at Draper, specializing in signal and power integrity for high-speed, impedance-controlled designs. Takes boards from schematic and layout through lab bring-up, building on radar integration at Raytheon and production hardware shipped at Tesla.

PROFESSIONAL EXPERIENCE

DRAPER
Cambridge, MA

Hardware Engineer MTS2, Harsh Environments Electronics — Apr 2026 – Present

- Own space-grade flight computer designs: requirements, schematic, design reviews, and documentation
- Define stack-ups and impedance/length-matching rules for SpaceWire, LVDS, and Ethernet; review layouts
- Simulate signal and power integrity (DC IR drop, AC PDN impedance) in HyperLynx, SIwave, and HFSS
- Bring up boards in the lab: power-rail validation and TDR/VNA impedance and S-parameter measurements
- Design for radiation and vibration: rad-tolerant parts, SEE mitigation, derating, and worst-case analysis
- Partner with mechanical packaging on thermal/vibe constraints and with firmware/software teams on bring-up

RAYTHEON
Woburn, MA

Electrical Engineer P2, Receiver Exciter & Back End Processing — Jan 2025 – Mar 2026

- Led receiver/exciter hardware & software integration on a $2.2B radar program, root-causing design flaws
- Validated 512 system design requirements by executing a custom acceptance test procedure
- Implemented up/down-converter PCBA changes improving signal-integrity margin and manufacturability

REDWIRE SPACE
Marlborough, MA

Hardware and Embedded Software Engineering Intern — May 2024 – Aug 2024

- Designed a low-cost analog sun sensor for LEO satellites, from concept to delivery in under 4 months
- Automated quadrant-photodiode calibration in Python and C++; validated the sensor on a solar simulator

LIBERTY DEFENSE
Wilmington, MA

Altium Design Engineering Consultant — May 2023 – June 2023

- Built unified Altium RF part library (25 impedance-controlled footprints), saving 350+ engineering hours/year

TESLA
Palo Alto, CA

Display Electrical Engineering Co-op — Jan 2022 – Aug 2022

- Redesigned 4 display PCBAs (schematic & layout) shipped in Model 3/Y/S/X, replacing end-of-life ICs
- Wrote firmware for new ICs and validated designs through EMI/EMC and environmental testing
- Streamlined mass-production PCBA validation, cutting validation time 15% and boosting throughput

FRESENIUS MEDICAL CARE
Lawrence, MA

R&D Electrical Engineering Co-op — Jan 2021 – Aug 2021

- Built an automated PCBA test platform; SPICE-modeled filters cut high-speed bus noise 75%

EDUCATION

NORTHEASTERN UNIVERSITY
Boston, MA

M.S. Electrical & Computer Engineering (4.0 GPA) – 2024
B.S. Electrical Engineering (Summa Cum Laude) – 2023

- Led 8 students on 3 Formula SAE EV projects; won 1st Place EV at IEEE Formula Hybrid+Electric 2021

TECHNICAL SKILLS

Hardware: Signal & Power Integrity, High-Speed Digital, Analog & RF Design, Board Bring-Up & Debug
Space: Rad-Tolerant Parts, SEE Mitigation, Derating, Worst-Case Analysis, IPC Class 3, NASA Workmanship
Tools: Xpedition, HyperLynx, SIwave, HFSS, Altium Designer, OrCAD, PSpice, LTspice, TDR/VNA
Software: C/C++, Python, MATLAB, Git, Jira, Confluence, AI/LLMs